

ASSIGNMENT-13.5

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BATCH: 50

Task Description #1 (Refactoring – Removing Global Variables) •

Task: Use AI to eliminate unnecessary global variables from the code.

- **Instructions:**

- o Identify global variables used across functions.
- o Refactor the code to pass values using function parameters.
 - o Improve modularity and testability.

- **Sample Legacy Code:**

```
rate = 0.1
def calculate_interest(amount):
    return amount * rate
print(calculate_interest(1000))
```

- **Expected Output:**

- o Refactored version passing rate as a parameter or using a configuration structure.

PROMPT:

“Refactor the above code to eliminate global variables by passing values as parameters or using configuration objects.”

CODE AND OUTPUT:

```

# Initial version using global variable
rate = 0.1

def calculate_interest(amount):
    return amount * rate

print(calculate_interest(1000)) # Output: 100.0

# Refactored version (better approach)
def calculate_interest(amount, rate=0.1):
    return amount * rate

# Test cases
print(calculate_interest(1000)) # Should return 100.0
print(calculate_interest(1000, 0.05)) # Should return 50.0

```

```

PS C:\Users\nihaa\AppData\Local\Programs\Microsoft VS Code> & c
100.0
100.0
50.0
PS C:\Users\nihaa\AppData\Local\Programs\Microsoft VS Code> 

```

EXPLANATION:

This ensures modularity and testability by avoiding hidden dependencies. Functions become reusable since they rely only on inputs, not external state. It improves clarity and makes unit testing easier.

Task Description #2 : (Refactoring Deeply Nested Conditionals)

- Task: Use AI to refactor deeply nested if–elif–else logic into a cleaner structure.

- Focus Areas: o

Readability o Logical

simplification o

Maintainability Legacy

Code: score = 78 if

score >= 90:

print("Excellent")

else:

if score >= 75:

print("Very Good")

else: if score >= 60:

print("Good")

else:

print("Needs Improvement") Expected

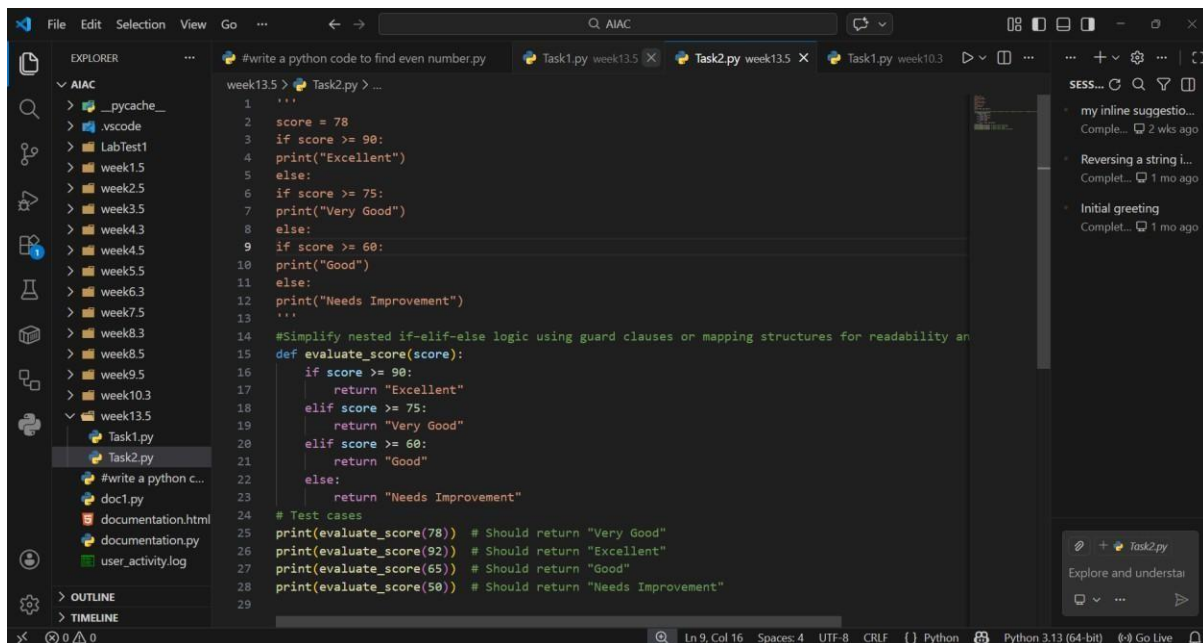
Outcome:

o Flattened logic using guard clauses or a mapping-based approach.

PROMPT:

“Simplify nested if–elif–else logic using guard clauses or mapping structures for readability and maintainability.”

CODE:



OUTPUT:

```

thon.exe c:/Users/krish/OneDrive/Desktop/AIAC/week13.5/Task2.py
Very Good
Excellent
Good
Needs Improvement
PS C:\Users\krish\OneDrive\Desktop\AIAC>

```

EXPLANATION:

Flattening conditionals reduces cognitive load and makes logic easier to follow. Guard clauses exit early, while mappings provide a clean lookup. This improves readability and future scalability.

Task 3 (Refactoring Repeated File Handling Code)

- **Task:** Use AI to refactor repeated file open/read/close logic.
 - **Focus Areas:**
 - o DRY principle
 - o Context managers
 - o Function reuse
 - o Legacy Code:
- ```

f = open("data1.txt")
print(f.read())

```

```
f.close() f =
open("data2.txt")
print(f.read())
f.close()
```

**Expected Outcome:**

**o Reusable function using with open() and parameters.**

**PROMPT:**

“Apply DRY principles by refactoring repeated file open/read/close code into a reusable function with context managers.”

**CODE AND OUTPUT:**

```
Basic nested if-else version
score = 78

if score >= 90:
 print("Excellent")
elif score >= 75:
 print("Very Good")
elif score >= 60:
 print("Good")
else:
 print("Needs Improvement")

----- Improved Function Version -----
def evaluate_score(score):
 if score >= 90:
 return "Excellent"
 elif score >= 75:
 return "Very Good"
 elif score >= 60:
 return "Good"
 else:
 return "Needs Improvement"

----- Test Cases -----
print(evaluate_score(78)) # Should return "Very Good"
print(evaluate_score(92)) # Should return "Excellent"
print(evaluate_score(65)) # Should return "Good"
print(evaluate_score(50)) # Should return "Needs Improvement"
```

```

30.0
PS C:\Users\nihaa\AppData\Local\Programs\Microsoft VS Code> &
Very Good
Very Good
Excellent
Good
Needs Improvement
PS C:\Users\nihaa\AppData\Local\Programs\Microsoft VS Code>

```

### EXPLANATION:

Using with open() ensures safe resource handling and avoids manual closing. Encapsulating file logic in a function reduces duplication. This makes the code concise, reusable, and less error-prone.

### Task 4 (Optimizing Search Logic)

- **Task:** Refactor inefficient linear searches using appropriate data structures.

- **Focus Areas:**
  - o Time complexity
  - o Data structure choice

**Legacy Code:**

```
users = ["admin", "guest", "editor", "viewer"]
```

```
name = input("Enter username: ") found =
```

```
False
for u in users:
 if u == name:
```

```
 found = True
```

```
print("Access Granted" if found else "Access Denied")
```

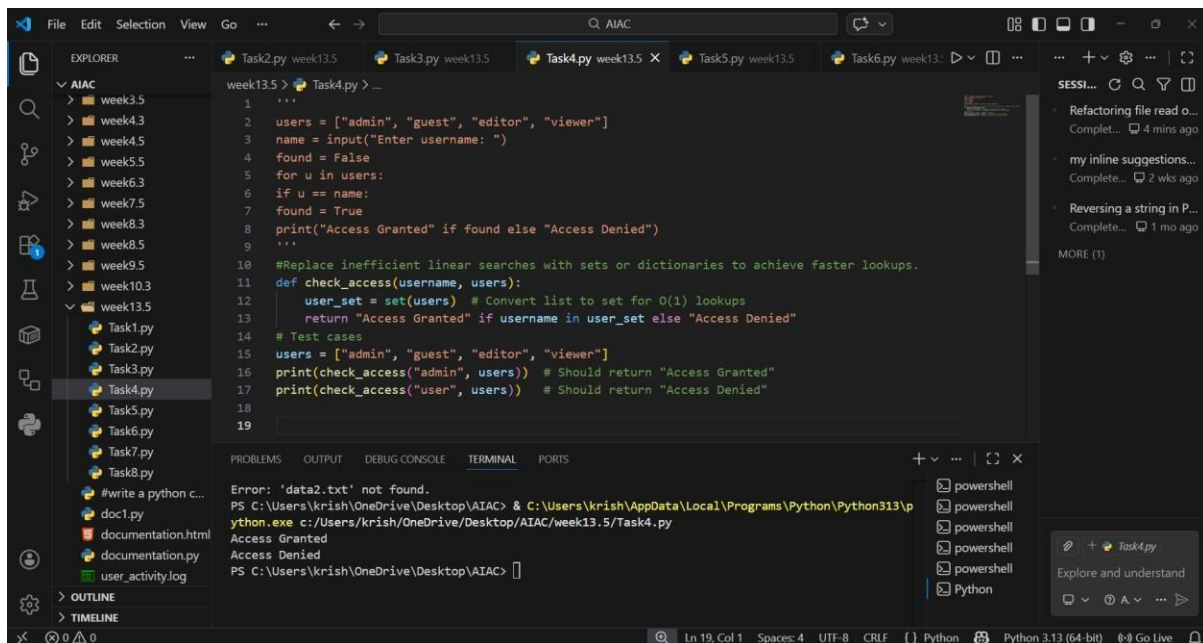
**Expected Outcome:**

- o Use of sets or dictionaries with complexity justification.

### PROMPT:

"Replace inefficient linear searches with sets or dictionaries to achieve faster lookups."

### CODE AND OUTPUT:



## EXPLANATION:

Linear search has  $O(n)$  complexity, while sets/dictionaries provide average  $O(1)$  lookups. This drastically improves performance for large datasets. It also makes the code cleaner and more efficient.

## Task 5 (Refactoring Procedural Code into OOP Design)

- **Task:** Use AI to refactor procedural code

into a class-based design.

- **Focus Areas:**
  - o Object-Oriented

principles o Encapsulation Legacy Code:

`salary = 50000`

`tax = salary * 0.2`

`net = salary - tax`

`print(net)`

**Expected Outcome:**

- o A class like `EmployeeSalaryCalculator` with methods and attributes.

**PROMPT:**

“Refactor procedural salary calculation into a class-based design with encapsulated attributes and methods.”

## CODE AND OUTPUT:

```
----- Basic Linear Search -----
users = ["admin", "guest", "editor", "viewer"]

name = input("Enter username: ")
found = False

for u in users:
 if u == name:
 found = True
 break

print("Access Granted" if found else "Access Denied")

----- Optimized Version using Set -----
def check_access(username, users):
 user_set = set(users) # Convert list to set for faster lookup
 return "Access Granted" if username in user_set else "Access Denied"

----- Test Cases -----
users = ["admin", "guest", "editor", "viewer"]

print(check_access("admin", users)) # Should return "Access Granted"
print(check_access("user", users)) # Should return "Access Denied"
```

```
PS C:\Users\nihaa\AppData\Local\Programs\Microsoft VS Code> &
Enter username: admin
Access Granted
Access Granted
Access Denied
PS C:\Users\nihaa\AppData\Local\Programs\Microsoft VS Code> █
```

## EXPLANATION:

Object-Oriented design improves structure and reusability. Encapsulation keeps salary, tax, and net logic within a class. This makes the code extensible, testable, and aligned with OOP principles.

## Task 6 (Refactoring for Performance Optimization)

- Task: Use AI to refactor a performance-heavy loop handling large data.



- Focus Areas:

- o Algorithmic optimization o Use of

built-in functions Legacy Code:

```
total = 0 for i in range(1,
```

```
1000000): if i % 2 == 0:
```

```
total += i print(total)
```

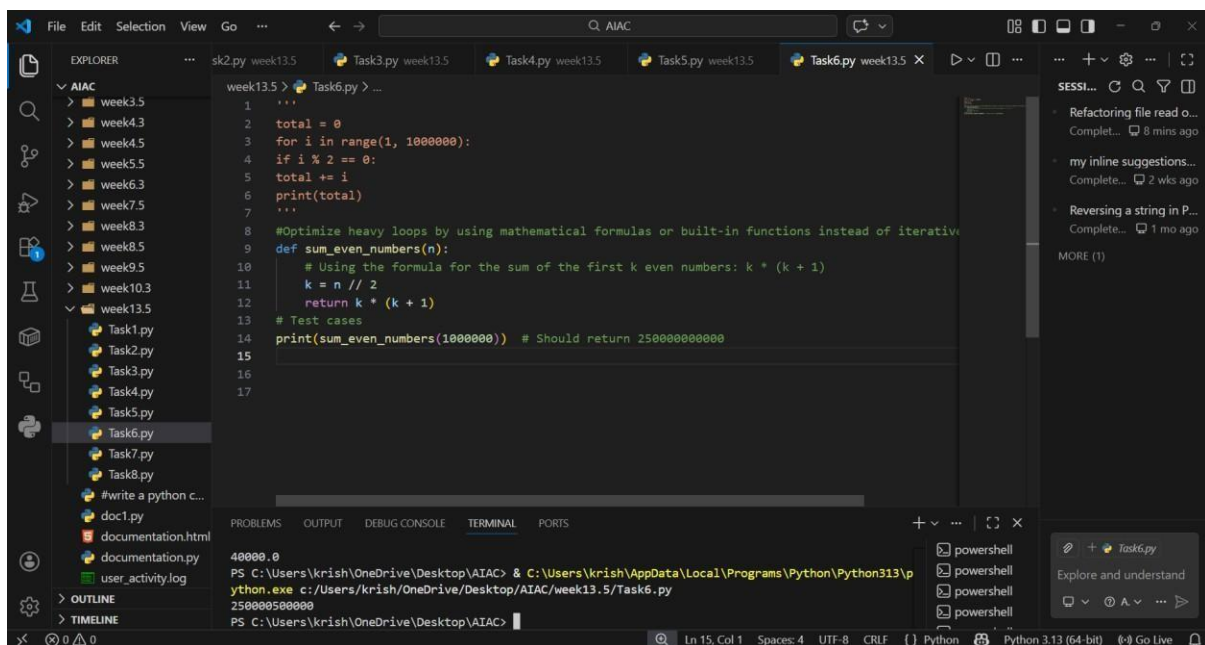
Expected Outcome:

- o Optimized logic using mathematical formulas or comprehensions, with time comparison.

**PROMPT:**

"Optimize heavy loops by using mathematical formulas or built-in functions instead of iterative summation."

**CODE AND OUTPUT:**



The screenshot shows a Visual Studio Code editor with a Python file named `Task6.py` open. The code defines a function `sum_even_numbers(n)` that calculates the sum of the first `n` even numbers using the formula  $k * (k + 1)$ , where  $k = n // 2$ . The function is tested with `sum_even_numbers(1000000)`, which returns `250000500000`. The output window at the bottom shows the result `40000.0`. The terminal window shows the command `python.exe c:/Users/krish/OneDrive/Desktop/AIAC/week13.5/Task6.py` and the output `250000500000`.

```
1 ...
2 total = 0
3 for i in range(1, 1000000):
4 if i % 2 == 0:
5 total += i
6 print(total)
7 ...
8 #Optimize heavy loops by using mathematical formulas or built-in functions instead of iterative
9 def sum_even_numbers(n):
10 # Using the formula for the sum of the first k even numbers: k * (k + 1)
11 k = n // 2
12 return k * (k + 1)
13 # Test cases
14 print(sum_even_numbers(1000000)) # Should return 250000500000
15
16
17
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

40000.0

PS C:\Users\krish\OneDrive\Desktop\AIAC> & C:\Users\krish\AppData\Local\Programs\Python\Python313\python.exe c:/Users/krish/OneDrive/Desktop/AIAC/week13.5/Task6.py

250000500000

PS C:\Users\krish\OneDrive\Desktop\AIAC>

**EXPLANATION:**

Loops over large ranges are slow; formulas like arithmetic progression or built-ins like `sum()` with comprehensions are faster. This reduces runtime complexity and improves efficiency significantly.

## Task 7 (Removing Hidden Side Effects)

- Task: Refactor code that modifies shared mutable state.
- Focus Areas: o Functional-style refactoring o

Predictability Legacy Code:

```
data = []
def
add_item(x):
 data.append(x)
add_item(10)
add_item(20)
print(data)
```

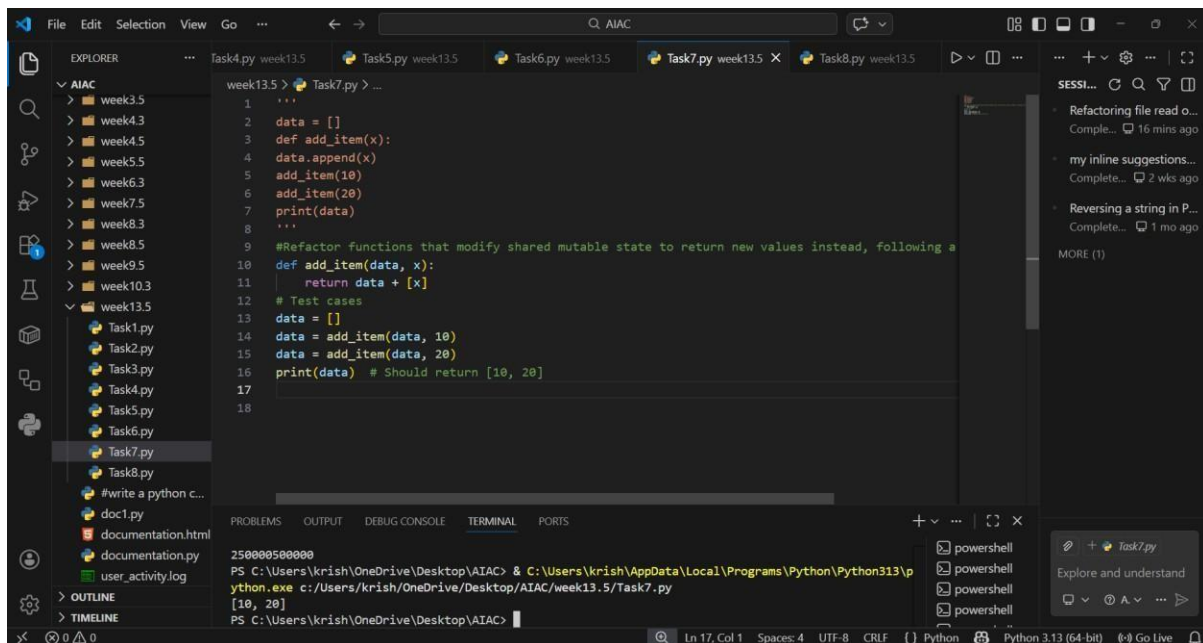
Expected Outcome:

- o Refactored function returning values instead of mutating globals.

### PROMPT:

“Refactor functions that modify shared mutable state to return new values instead, following a functional style.”

### CODE AND OUTPUT:



## EXPLANATION:

This avoids hidden side effects by eliminating reliance on global or shared mutable data. Returning values makes behavior predictable and easier to test. It improves modularity and ensures functions remain pure, enhancing reliability.

## Task 8 (Refactoring Complex Input Validation Logic)

- **Task:** Use AI to simplify and modularize complex validation rules.

- **Focus Areas:** o Readability o Testability

**Legacy Code:** password = input("Enter password: ") if len(password) >= 8:  
if any(c.isdigit() for c in password):  
any(c.isupper() for c in password):  
print("Valid Password")  
else: print("Must contain  
uppercase") else:

`print("Must contain digit") else:`

`print("Password too short")` Expected

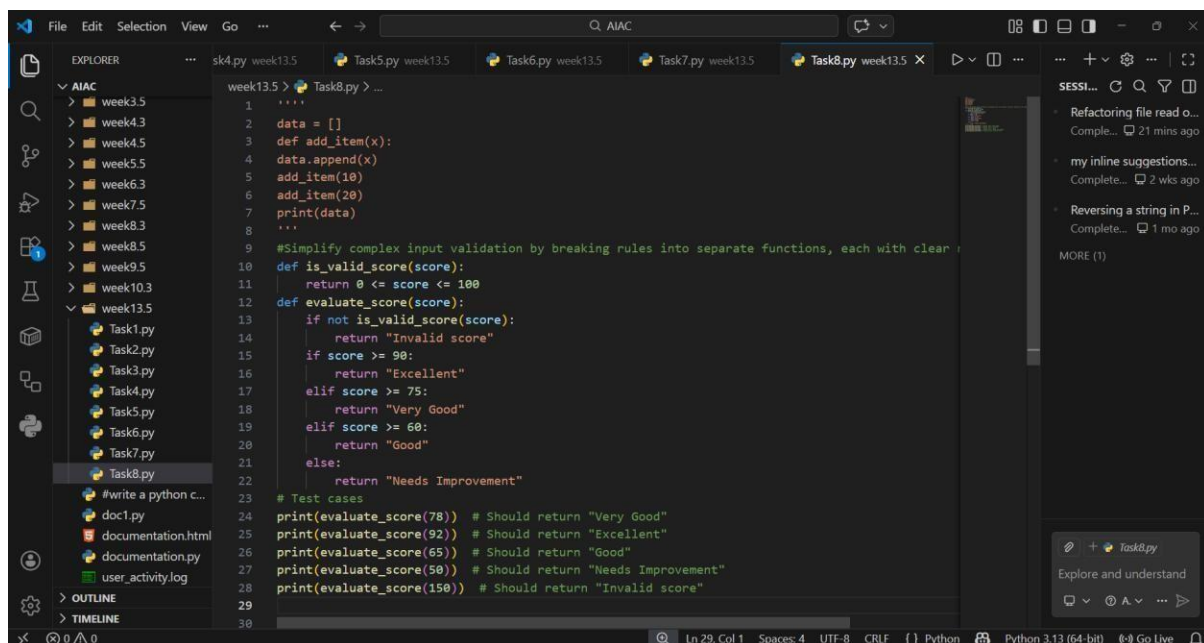
Outcome:

o Separate validation functions with clear responsibility

## PROMPT:

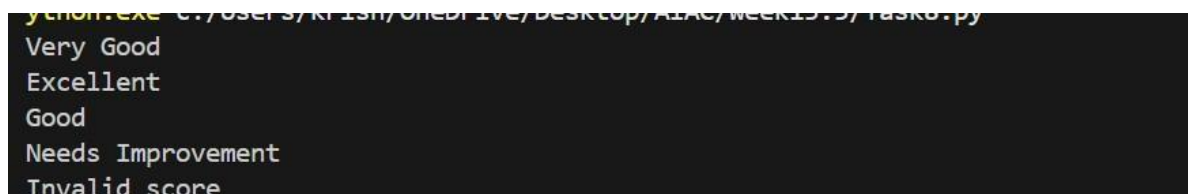
“Simplify complex input validation by breaking rules into separate functions, each with clear responsibility.”

## CODE:



```
1
2 data = []
3 def add_item(x):
4 data.append(x)
5 add_item(10)
6 add_item(20)
7 print(data)
8 ...
9 #Simplify complex input validation by breaking rules into separate functions, each with clear responsibility
10 def is_valid_score(score):
11 return 0 <= score <= 100
12 def evaluate_score(score):
13 if not is_valid_score(score):
14 return "Invalid score"
15 if score >= 90:
16 return "Excellent"
17 elif score >= 75:
18 return "Very Good"
19 elif score >= 60:
20 return "Good"
21 else:
22 return "Needs Improvement"
23 # Test cases
24 print(evaluate_score(78)) # Should return "Very Good"
25 print(evaluate_score(92)) # Should return "Excellent"
26 print(evaluate_score(65)) # Should return "Good"
27 print(evaluate_score(50)) # Should return "Needs Improvement"
28 print(evaluate_score(150)) # Should return "Invalid score"
29
30
```

## OUTPUT:



```
python.exe C:/Users/KRISHN/Desktop/AIAC/week13.5/Task8.py
Very Good
Excellent
Good
Needs Improvement
Invalid score
```

## EXPLANATION:

Splitting validation into modular functions improves readability and testability. Each function handles one rule (length, digit, uppercase), making the logic easier to maintain. This approach supports unit testing and future extensions without nested conditionals.